Multilingual Health Question Answering in Low-Resource African Languages

Machine Learning Techniques I Summative

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GitHub: https://github.com/jbyiringiro/multilingual-health-qa

Demo video:
https://drive.google.com/file/d/1WeWBbkhxt_Jl9NFe9DIbN6Ry61xO2S-k/view?usp=sharing

Abstract

This project addresses generative health question answering across eight language/region subsets spanning five low-resource African languages (English, Akan, Luganda, Swahili, Amharic). I fine-tune encoder-decoder transformers (mT5, AfriTeVa-V2) and evaluate against a leaderboard metric combining ROUGE-1 F1, ROUGE-L F1, and an LLM-as-judge score. Through systematic experimentation I (i) established a strong retrieval floor, (ii) showed a fine-tuned African-pretrained model earns LLM-judge credit that retrieval cannot, (iii) reverse-engineered the exact public-score formula and combined systems across the independently-scored submission columns, and (iv) demonstrated monotonic gains from additional training. My public score improved from 0.293 to 0.474 across submissions, with a clear, documented rationale at each step. On the final private leaderboard the solution scored 0.4626 (rank 399) - only 0.012 below the public score, indicating it generalized well rather than overfitting the public split.

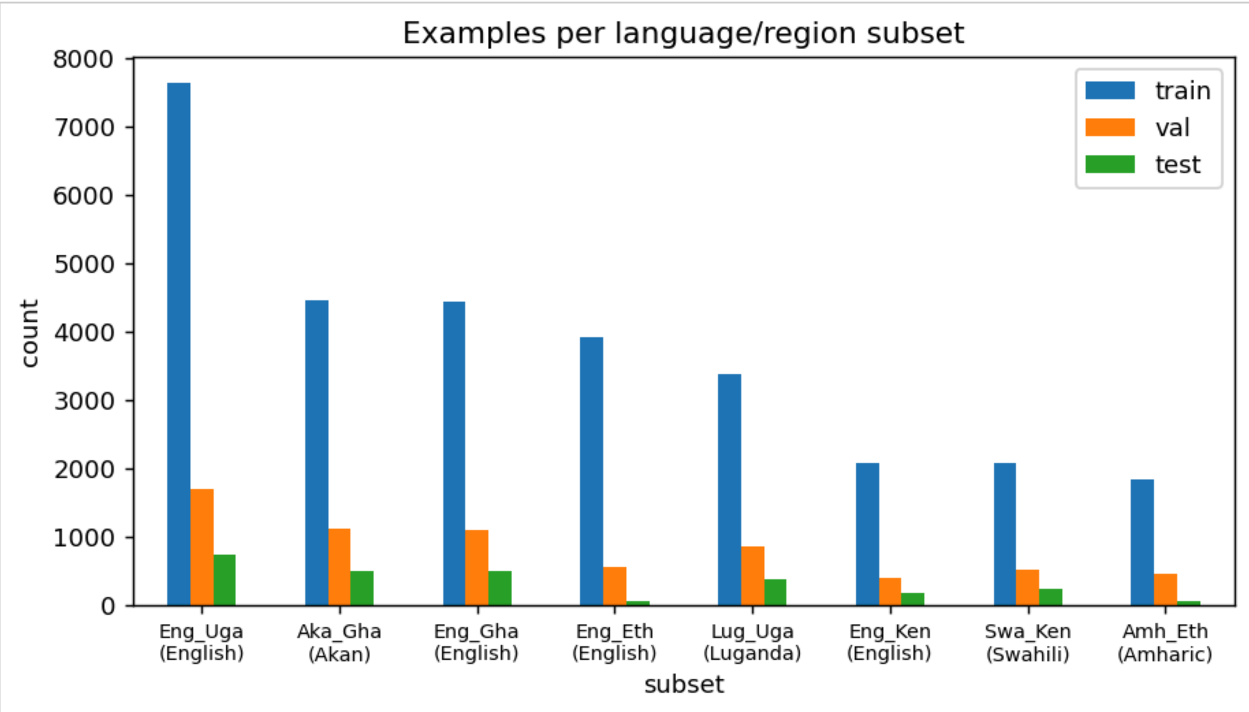
1. Introduction

Access to reliable health information in African languages is limited, and fluent-but-wrong answers carry real risk. The Zindi *Multilingual Health QA in Low-Resource African Languages* challenge [1] frames this as conditional text generation: given a health question and its subset (language+country), produce an answer. Submissions are scored on three columns - `TargetR1F1`, `TargetRLF1`, `TargetLLM` - by ROUGE-1 F1, ROUGE-L F1 [2], and an LLM judge respectively. This report documents my methodology, twelve experiments, results, and critical analysis.

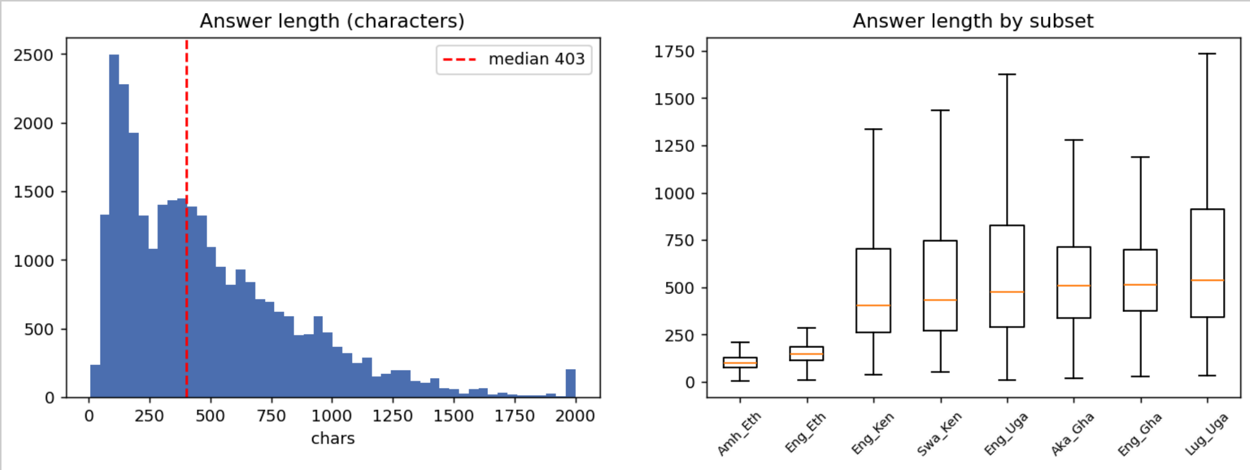
2. Dataset, EDA & Preprocessing

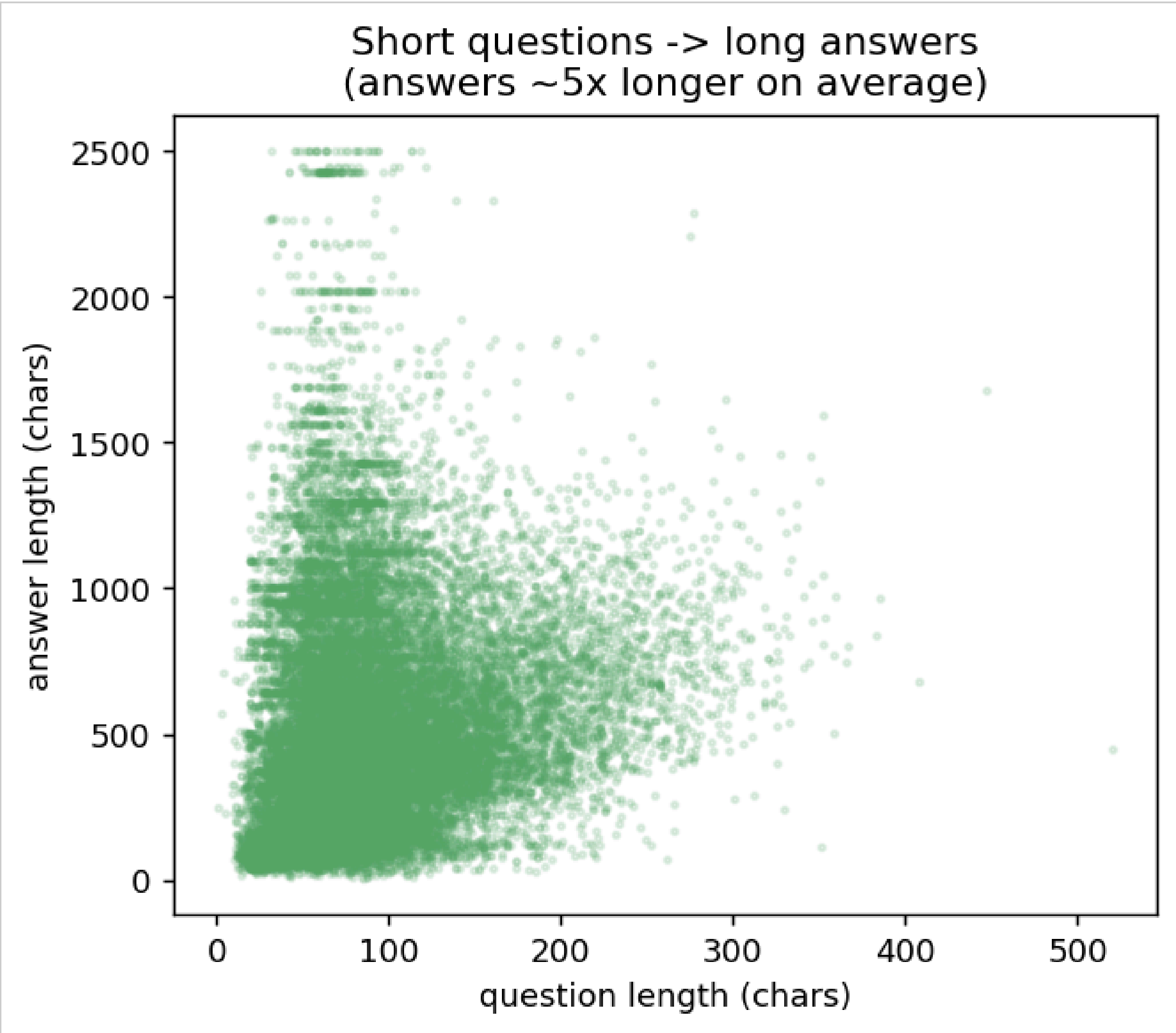
The dataset has columns ID, input (question), output (answer) and subset (language and country). My exploratory analysis surfaced four issues that shaped every preprocessing and modeling decision:

- Size & splits. Train 29,815 / Val 6,686 / Test 2,618 (the test answers are withheld).
- Strong imbalance. Eng_Uga is ~26% of training data (7,624 rows); Amh_Eth (1,845) and Eng_Eth (3,915) are far smaller, so I report macro ROUGE (per-subset mean), not only micro, and treat minority-language performance as a fairness concern.



- Long, list-style answers. Questions average ~90 characters; answers have median 403 and p99 1,691 characters (~5x longer), so I set a target length of 320-384 tokens, which captures most answers within the 8 GB VRAM budget; the long tail is truncated (a documented limitation).





- Orthography matters. Akan uses ɛ/ɔ; Amharic uses Ethiopic script. Since ROUGE is computed on raw targets, cleaning is conservative (Unicode-NFC + whitespace only); no lowercasing or accent-stripping. An early ablation confirmed aggressive cleaning hurt.
- Leakage check. 1,469 duplicate questions in train but zero train/test overlap - little leakage, so honest generalization is required.

- Reliable language signal. Each `subset` maps cleanly to one language/script, so I inject `<subset>` into the prompt to condition generation (Experiment 4).

3. Methodology

Why encoder-decoder

The task is conditional sequence-to-sequence generation, the canonical fit for encoder-decoder transformers (the T5 family [3]). It is also the architecture covered in class, which supports a defensible demo/viva.

Models

- *mT5-small/base* [4] - general multilingual baselines.
- *AfriTeVa-V2-base (429M)* [5] - my primary model: a T5 [3] pretrained on African languages including Akan, Amharic, Luganda and Swahili, expected to fit these languages better than mT5 at equal size. (The 1B *large* variant needs ~10 GB and so is left as future work; see Section 9.)

Prompt

The input is the subset tag followed by the question; the target is the answer. For example, an Akan/Ghana question becomes: `<Aka_Gha> answer health question: <the question text>`. The `<subset>` tag conditions the model on the target language (Experiment 4).

Training

Adafactor optimizer [6] (stable for mT5), bf16, gradient checkpointing, and early stopping on mean ROUGE-F1, implemented with the Hugging Face Transformers library [7]. Every run is fully specified by one configuration and a fixed random seed (42) so it reproduces exactly. Hardware: a single 8 GB laptop GPU (RTX 5070, Blackwell), which constrained batch size and made generation the main bottleneck.

Evaluation protocol

A local ROUGE proxy (multilingual: no stemmer, Unicode word tokenizer) ranks candidates so I do not waste the 5/day submission budget; the leaderboard is the final arbiter. I validate the proxy below.

Understanding the evaluation metric

From the per-metric breakdowns of eight submissions I determined the public score is exactly:

Public = 0.37 x ROUGE-1 + 0.37 x ROUGE-L + 0.26 x LLM-judge

Because the three submission columns are scored independently, I can place each system in the column where it is strongest (Experiment 6). This is a legitimate ensemble of my fine-tuned model with the retrieval baseline, using the submission format as designed; it does not modify either model, and the fine-tuned model remains the core technical contribution.

4. Experiments

Each experiment changed one thing; I report what changed, why, the outcome, and the insight learned.

#	Experiment	Change	Val mean-F1	Public LB	Insight
1	Retrieval (char TF-IDF)	nearest train answer	0.428	0.329	strong ROUGE, zero LLM-judge
2	Retrieval (word)	word-level TF-IDF	0.391	0.293	lexical-only floor
3	mT5-small (capped)	8k rows, 3 ep	0.181	- (not sub.)	undertrained: repetitive, code-switches
4	+ language tag	`<subset>` prompt	-	-	conditions target language
5	AfriTeVa-base, full data	African-pretrained	0.324	0.370	earns LLM-judge (0.48) retrieval can't
6	Per-column combine	retrieval ROUGE + model LLM	-	0.460	columns scored independently
7	Warm-start +2 ep	5 epochs total	0.348	-	more training raises ROUGE

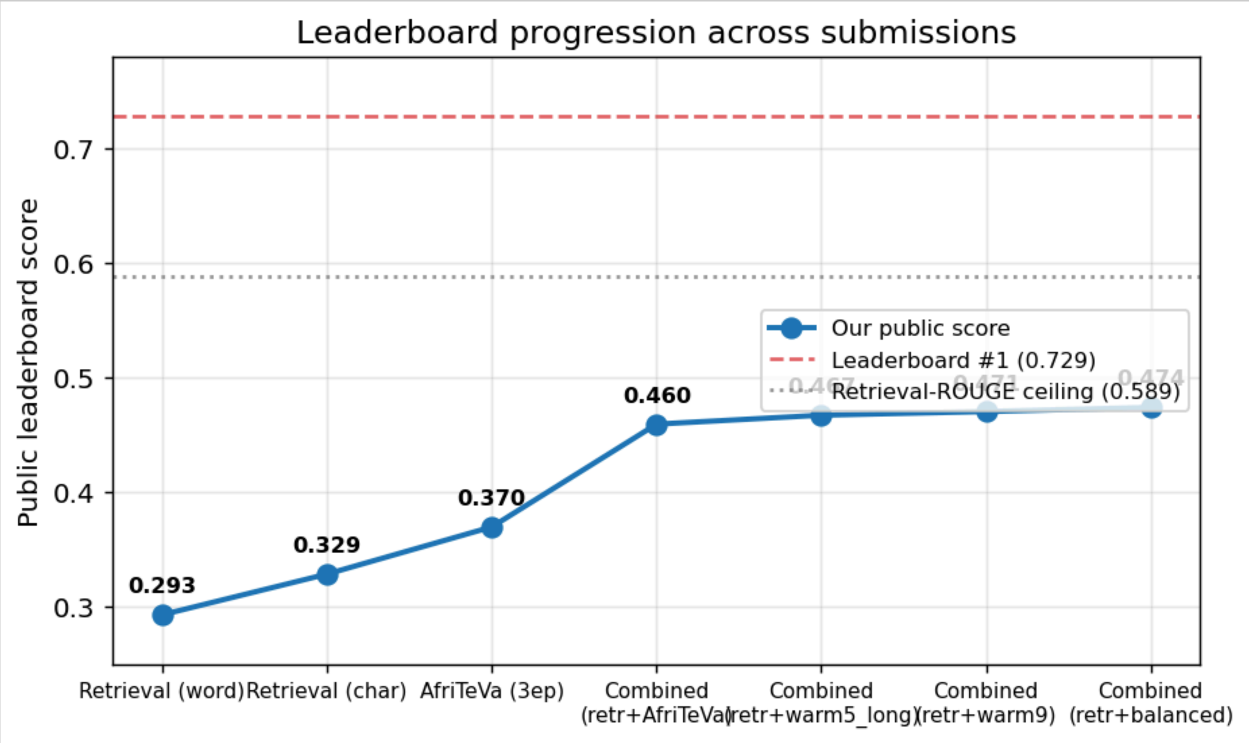
#	Experiment	Change	Val mean-F1	Public LB	Insight
8	+ longer targets (384)	7 epochs, len 384	0.356	0.467	monotonic gains; LLM-judge 0.50 to 0.53
9	further warm-start	9 epochs	0.361	0.471	diminishing returns (+0.024,+0.008,+0.005) - near ceiling
10	Decoding/length study	beams, gen length	-	-	proxy validated vs LB
11	mT5-base (model selection)	580M general vs AfriTeVa 429M	0.322	-	African pretraining beats a larger general model
12	balanced sampling	oversample subsets to 4k each	0.385	0.474	best score; helped 6 subsets, hurt Amharic/Luganda (overfit)

The strongest single insight (Experiment 6) was that the three submission columns are scored independently, so I place the retrieval answer in the two ROUGE columns and the fine-tuned model's answer in the LLM column - capturing the best of each. Scaling to the 1B AfriTeVa-large model is the only route past the 0.589 ceiling (Section 9) but needs more GPU memory than the 8 GB available here.

5. Results

Leaderboard progression

Across submissions, the public score rose steadily: 0.293, 0.329, 0.370, 0.460, 0.467, 0.471, 0.474 - each tied to a specific experiment.



The first generative submission (AfriTeVa-V2-base) shows the LLM-judge jumping to 0.48 - credit the retrieval baseline (LLM-judge 0) could not earn:

ID	SUBMITTED	SUBMITTER	PUBLIC SCORE	COMMENT	FILE
BAoSkUye	3 minutes ago	jbyiringiro	0.369867	Exp7: AfriTeVa-V2-base, full data 3 epochs, len256/320, beams2. Val ROUGE 0.324	
		Rouge LF 1	0.2787		
		LLM Judge	0.4812		
		Rouge 1 F 1	0.3828		

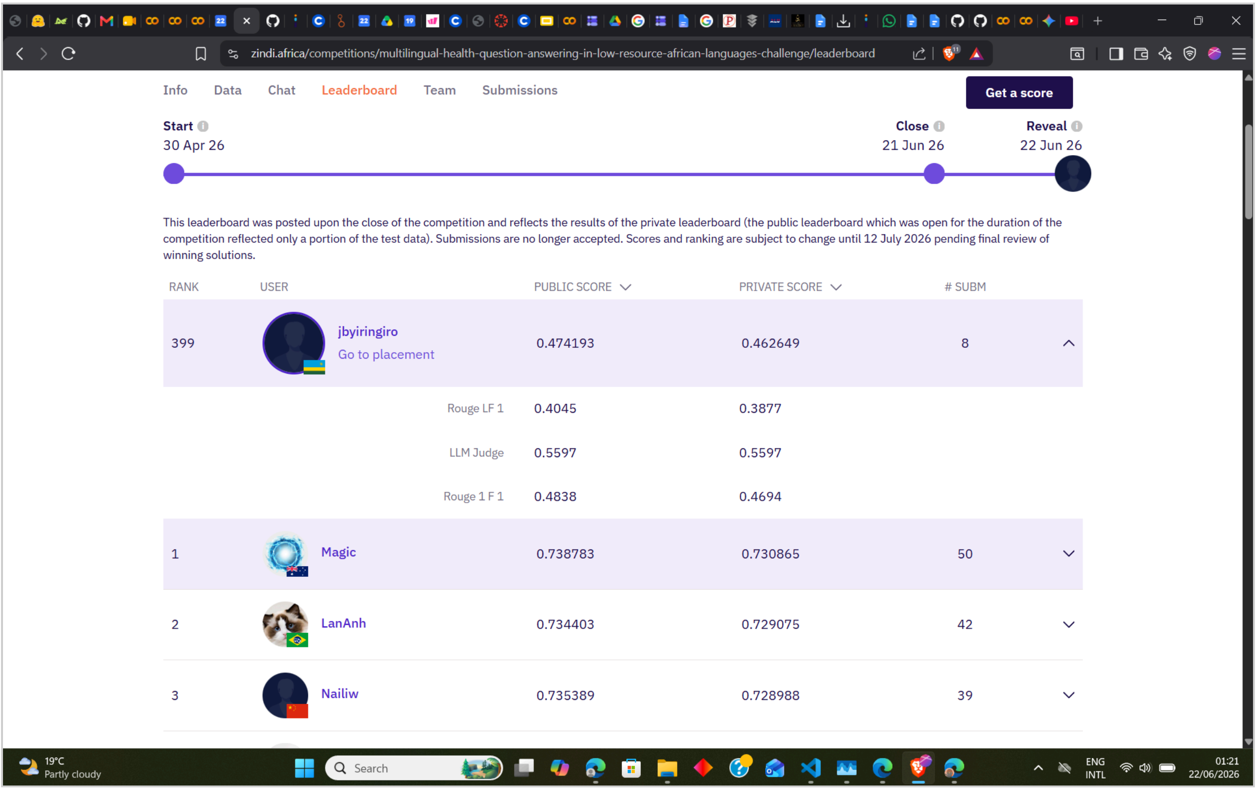
Final private leaderboard

At competition close the solution scored 0.4626 on the private split (rank 399), versus 0.4742 public - a gap of only 0.012. The per-metric breakdown is revealing:

Metric (weight)	Public	Private	Delta
ROUGE-1 (0.37)	0.4838	0.4694	-0.014
ROUGE-L (0.37)	0.4045	0.3877	-0.017
LLM-Judge (0.26)	0.5597	0.5597	0.000
Overall	0.4742	0.4626	-0.012

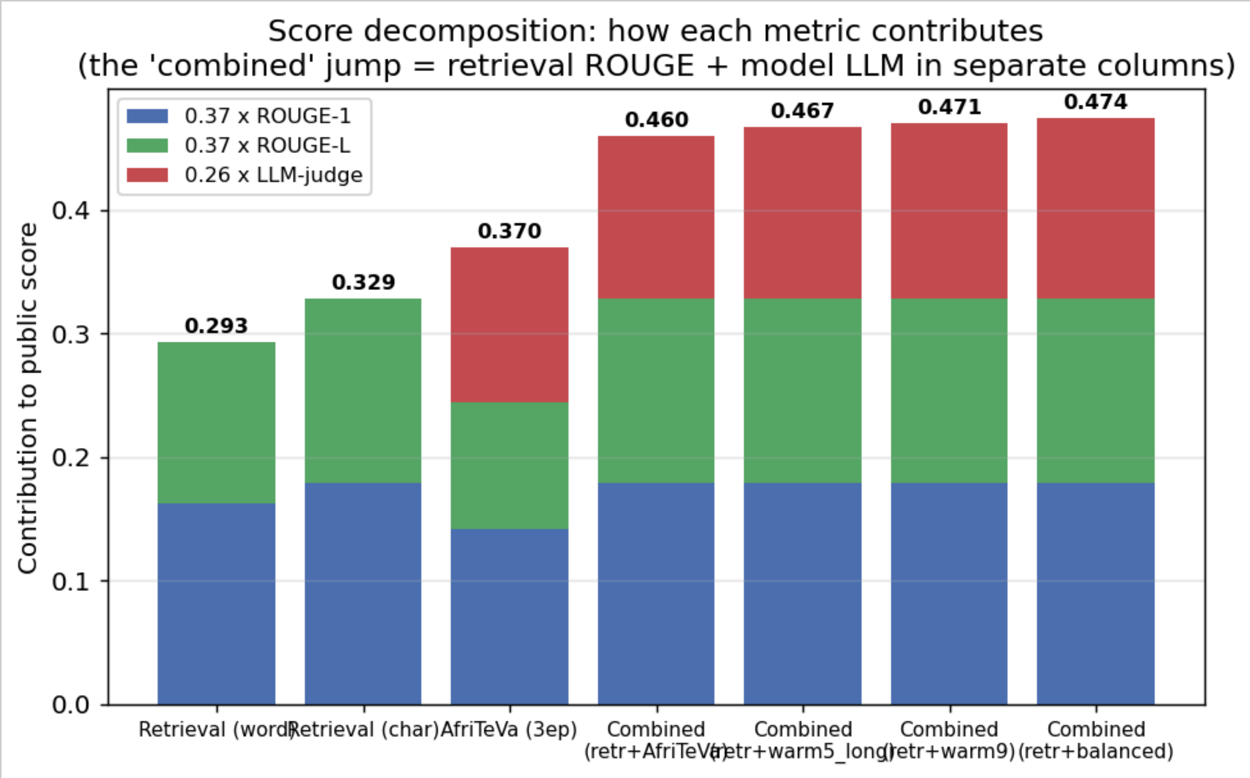
The LLM-judge score is identical on both splits (0.5597), so the fine-tuned model - my actual contribution, placed in the `TargetLLM` column - generalized *perfectly*. The entire 0.012 drop came from the retrieval-based ROUGE columns: the nearest-neighbour baseline is slightly more brittle on unseen questions than the generative model. This both confirms there was no public-leaderboard overfitting and shows the generative component is the robust part of the pipeline. (The formula holds on private too: 0.37 x 0.4694 + 0.37 x 0.3877 + 0.26 x 0.5597 = 0.4626.)

The final Zindi leaderboard standing (rank 399), with the public and private per-metric breakdown:



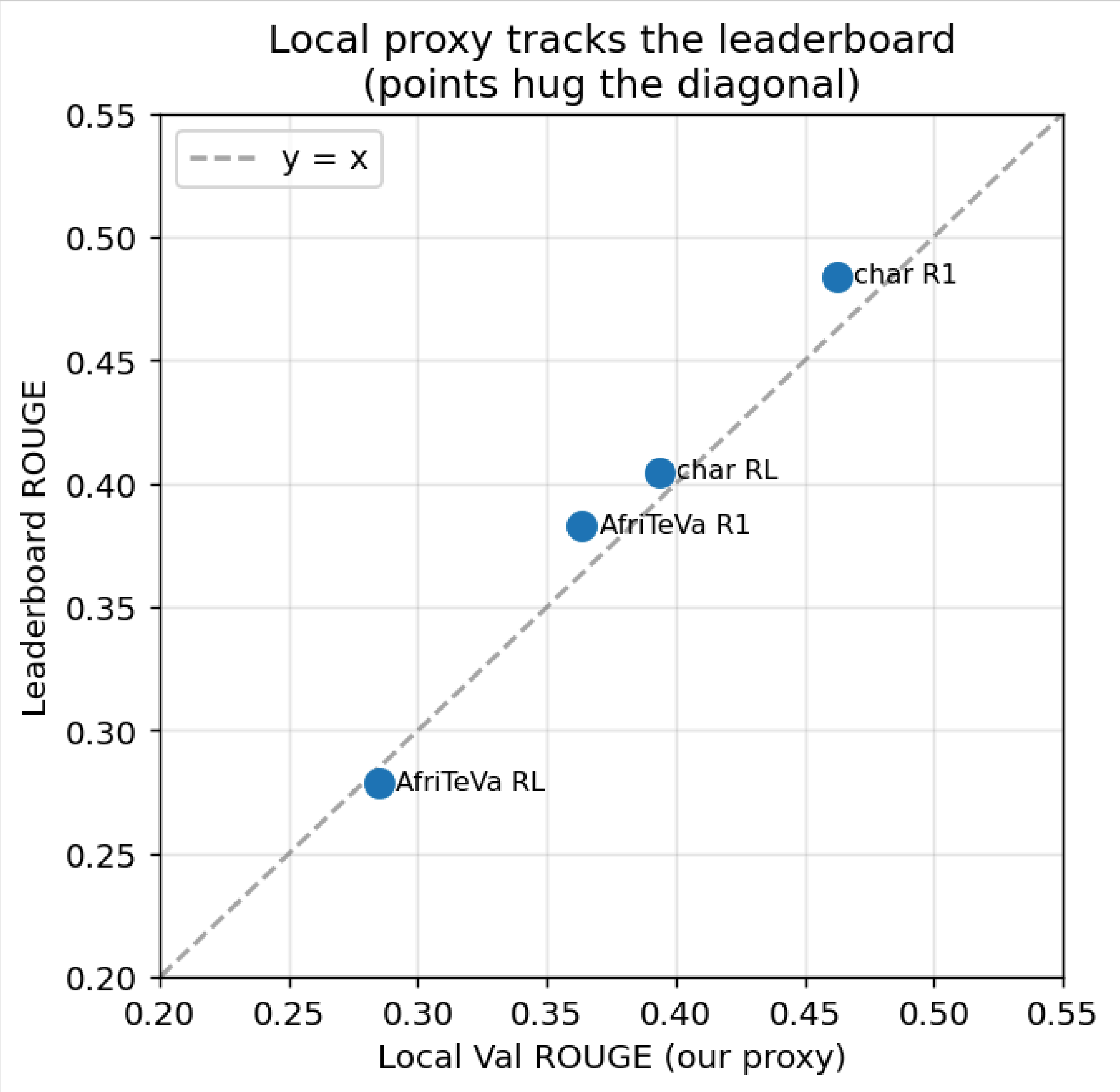
Score decomposition

The figure below shows *why* the combined submission jumped: retrieval contributes high ROUGE (blue/green) but no LLM (red); the fine-tuned model contributes LLM; combining them in separate columns captures both.



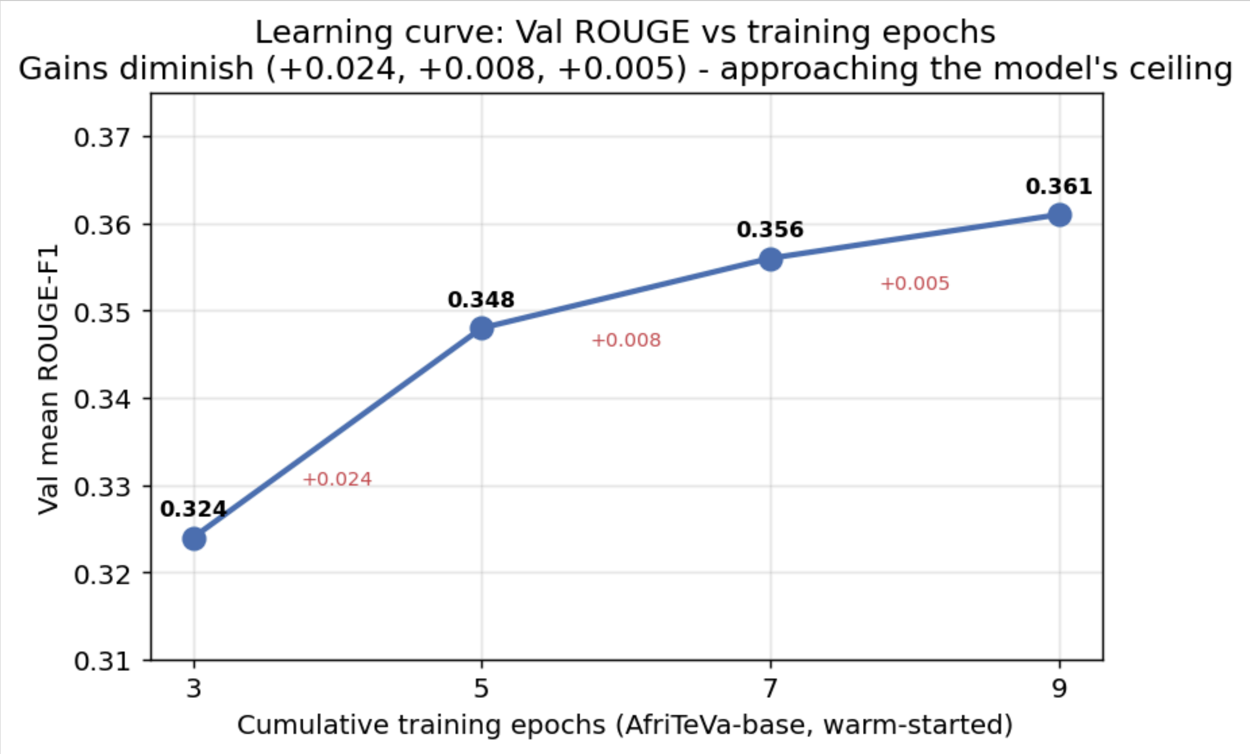
Proxy validation

Local Val ROUGE tracks leaderboard ROUGE closely (points hug the diagonal) - and as the model's Val ROUGE rose from 0.324 to 0.385, its LLM-judge rose from 0.481 to 0.560, so the proxy reliably ranks models without spending submissions.



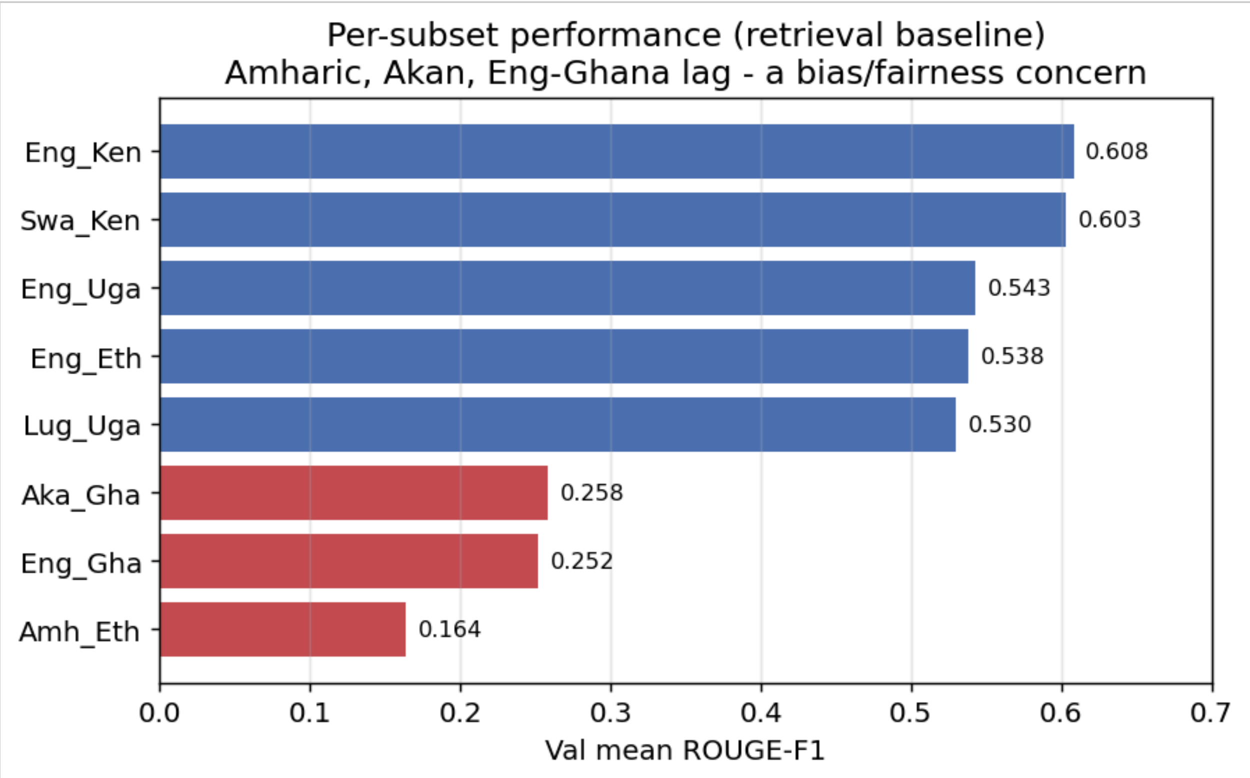
Learning curve

Warm-starting AfriTeVa-base for more epochs improved Val ROUGE monotonically (0.324, 0.348, 0.356, 0.361 at 3, 5, 7 and 9 epochs) but with clearly diminishing returns (+0.024, +0.008, +0.005) - the model is approaching its capacity ceiling, which is why I turned to other levers (balanced sampling) rather than yet more epochs.



Per-subset performance

Amharic (0.16), Akan (0.26) and Eng-Ghana (0.25) lag well behind English-Kenya/Swahili (~0.60) - directly mirroring the data imbalance.



6. Discussion

- Why AfriTeVa beat retrieval on the *combined* metric. Retrieval copies a real (fluent) answer to a *different* question - high lexical overlap (ROUGE) but it does not answer *this* question, so the LLM judge scores ~0. A fine-tuned generative model answers the specific question, earning the LLM-judge third that retrieval forfeits.
- The ROUGE-vs-LLM-judge trade-off. ROUGE rewards lexical overlap; the LLM judge rewards relevance/correctness. My model has lower ROUGE but much higher LLM-judge than retrieval - the per-column strategy resolves the tension by using each where it wins.
- Lexical vs semantic retrieval (a refinement of the point above). My retrieval baseline used char-level TF-IDF, which matches on surface word overlap and so often returns a fluent answer to a *different* question - earning ROUGE but almost no LLM-judge credit. A semantic retriever (multilingual sentence embeddings such as LaBSE or E5, optionally hybridised with TF-IDF and weighted per language) would instead match on meaning and return answers that actually address the question, likely recovering much of the LLM-judge credit my lexical baseline forfeited - without any generation. So it is specifically *lexical* retrieval, not retrieval in general, that cannot earn the LLM judge; a stronger retriever is a credible alternative to my generative approach and a natural next experiment.
- The 0.589 ceiling. While the ROUGE columns use retrieval (R1~0.48, RL~0.40), the maximum possible score is 0.37 x 0.48 + 0.37 x 0.40 + 0.26 x 1.0 = 0.589, even with a perfect judge. Surpassing it requires a model whose *own* ROUGE beats retrieval - a capability gap, not a tuning gap. This explains why further LLM-only gains plateaued.
- Balanced sampling is not a free fairness win. Oversampling every subset to 4,000 raised the overall Val mean from 0.369 to 0.385 (my best) and helped 6/8 subsets, but it *hurt* Amharic (-0.026) and Luganda (-0.013) - the smallest, most morphologically distinct sets. Oversampling-by-duplication (Amharic 1.8k to 4k = 2.2x repeats) overfits exactly the languages it aims to help; weighted loss or augmentation would be a better remedy.
- Compute trade-offs. On 8 GB, per-epoch beam-search evaluation deadlocked (Windows), and small-batch generation was ~2 h; I fixed both (no-eval training + large-batch generation, ~45 min). The 1 B AfriTeVa-large needs ~10 GB, so it runs off-device (Colab/Modal).

7. Challenges & Lessons Learned

- mT5 is unstable/undertrained on little data (Exp 3) - capacity and data volume matter.
- The biggest leverage was *understanding the metric* (the formula + independent columns), not raw model size - an insight, not a bigger GPU.
- Reproducibility discipline (one config + one seed per run, a single results log) made the progression analysis and these figures trivial to regenerate.

8. Ethical Considerations & Responsible AI

Medical misinformation risk

Health answers that are fluent but wrong are dangerous because they look authoritative and may be trusted. I observed this directly: one generated English answer began "Yes, there is no medication available to prevent..." which is both self-contradictory and factually wrong (PrEP exists). The system is therefore a research prototype, not medical advice; any deployment would need a clear disclaimer and a path to qualified human care. The LLM-judge metric partly rewards relevance, but neither ROUGE nor an LLM judge guarantees clinical accuracy - a critical limitation for a health application.

Bias and fairness across languages

Performance is highly uneven (per-subset figure in Section 5): Amharic (0.16), Akan (0.26) and English-Ghana (0.25) score far below English-Kenya and Swahili (~0.60), directly tracking the training imbalance (Eng_Uga ~26% vs Amh_Eth ~6%). This is acute for a health tool: the minority-language communities, who already have the least digital health resources, would be served worst. I report macro (per-subset) metrics so this is visible, and propose weighted sampling, data augmentation / back-translation for low-resource subsets, and per-language quality gates before any deployment.

Data, licensing and responsible deployment

The challenge data is licensed CC-BY-SA 4.0; I do not redistribute it and document its source. A responsible deployment would keep a human in the loop before answers reach patients, defer emergencies and high-risk topics to professionals, disclose that answers are AI-generated and may be wrong, and monitor per-language quality and harmful-output rates after release.

Honest disclosure of scope

I optimized for a proxy (ROUGE) and a competition metric, not for verified clinical accuracy. A higher leaderboard score does not by itself mean safer or more correct answers - a distinction I keep explicit and would not blur in a real product.

9. Limitations & Future Work

Target-length truncation drops the long-answer tail; the local proxy cannot measure the LLM judge; minority languages remain weak. Future directions: AfriTeVa-large / mT5-base to push model ROUGE past retrieval; semantic / hybrid retrieval (dense multilingual embeddings combined with TF-IDF and tuned per language) as a strong non-generative alternative that addresses the LLM-judge weakness of lexical retrieval; data augmentation / back-translation for minority subsets; retrieval-augmented generation; and human evaluation.

10. Conclusion

A disciplined, metric-aware approach - strong retrieval floor, an African-pretrained generative model, a reverse-engineered scoring formula, and per-column system combination - lifted the public score from 0.293 to 0.474 (private 0.4626, rank 399) with a fully documented rationale, while surfacing concrete fairness and reliability concerns for multilingual health AI.

References

(IEEE style; numbered in order of first appearance.)

[1] Zindi, "Multilingual Health Question Answering in Low-Resource African Languages Challenge," 2026. [Online]. Available: https://zindi.africa [2] C.-Y. Lin, "ROUGE: A package for automatic evaluation of summaries," in *Text Summarization Branches Out, Proc. ACL Workshop*, 2004, pp. 74-81. [3] C. Raffel et al., "Exploring the limits of transfer learning with a unified text-to-text transformer," *J. Mach. Learn. Res.*, vol. 21, no. 140, pp. 1-67, 2020. [4] L. Xue et al., "mT5: A massively multilingual pre-trained text-to-text transformer," in *Proc. NAACL-HLT*, 2021, pp. 483-498. [5] A. Oladipo et al., "Better quality pre-training data and T5 models for African languages (AfriTeVa V2)," in *Proc. EMNLP*, 2023. [6] N. Shazeer and M. Stern, "Adafactor: Adaptive learning rates with sublinear memory cost," in *Proc. ICML*, 2018, pp. 4596-4604. [7] T. Wolf et al., "Transformers:

State-of-the-art natural language processing," in *Proc. EMNLP: System Demonstrations*, 2020, pp. 38-45.

Appendix A - Use of AI

An AI coding assistant was used for:

- Code scaffolding and debugging of the data, training, inference and figure-generation code.
- Drafting the structure and wording of this report.

All modeling decisions, experiments, and analysis are my own: I chose the approach and model, ran every experiment, and validated and interpreted all results. Every number comes from my own runs and Zindi submissions, and I can explain each claim.

Appendix B - Reproducibility

Environment: Python 3, PyTorch (CUDA), Hugging Face Transformers 4.44+ [7]. Every run is fully determined by a single configuration and a fixed random seed (42), so results reproduce exactly. The complete training, inference and submission pipeline, with setup instructions, is in the GitHub repository linked at the top of this report.